

SE Lab - 2

JIRA – AGILE PROJECT MANAGEMENT

Problem Statement #08

Digital Campus Library Reservation Gateway

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Branch: AIML

Section: D

1. Project Overview

Problem Statement: Digital Campus Library Reservation Gateway

The project focuses on developing a digital gateway for campus library operations. The system allows Student Members to search the physical ISBN catalog, place holds on loaned books, manage FIFO reservation queues, receive availability notifications, and supports the Head Librarian in managing overdue fines and administrative operations.

Actors

- Student Member
- Head Librarian

Jira Project -

Digital Campus Library Reservation Gateway

Methodology -

Agile Scrum

2. Epics and User Stories

Epic	User Story	Priority	Story Points
Catalog Discovery & Search	Search ISBN Catalog	High	3
	View Book Details	High	2
	View Paginated Search Results	Medium	3
Hold Request & Reservation	Place Hold Request	High	5
	Validate Hold Eligibility	High	3
	Display Queue Position	High	2
FIFO Queue Management	Add Member to FIFO Queue	High	3
	Update Queue Positions	Medium	5
	Fulfill Next Queue Reservation	Medium	5
Hold Availability Notifications	Send Hold Availability Notification	High	3
	Notify Next Eligible Member	Medium	3
Librarian Fine Management	Record Overdue Fine	Medium	5
	View Fine Status	Medium	3
Performance & Access Security	Fast Catalog Search	High	8

Restrict Librarian Operations	High	5
----------------------------------	------	---

Total

15 User Stories — 58 Story Points

3. Epic Creation and Backlog

The project was organized into six Epics representing the major functional and non-functional areas of the Digital Campus Library Reservation Gateway. User Stories were created under their respective Epics with appropriate priorities and story-point estimates.

4. User Story Details

Each User Story was created using the Agile user-story format and associated with its respective Epic. Priority and Story Points were assigned to support backlog planning and sprint estimation.

5. Sprint 1 – Planning

Sprint Goal

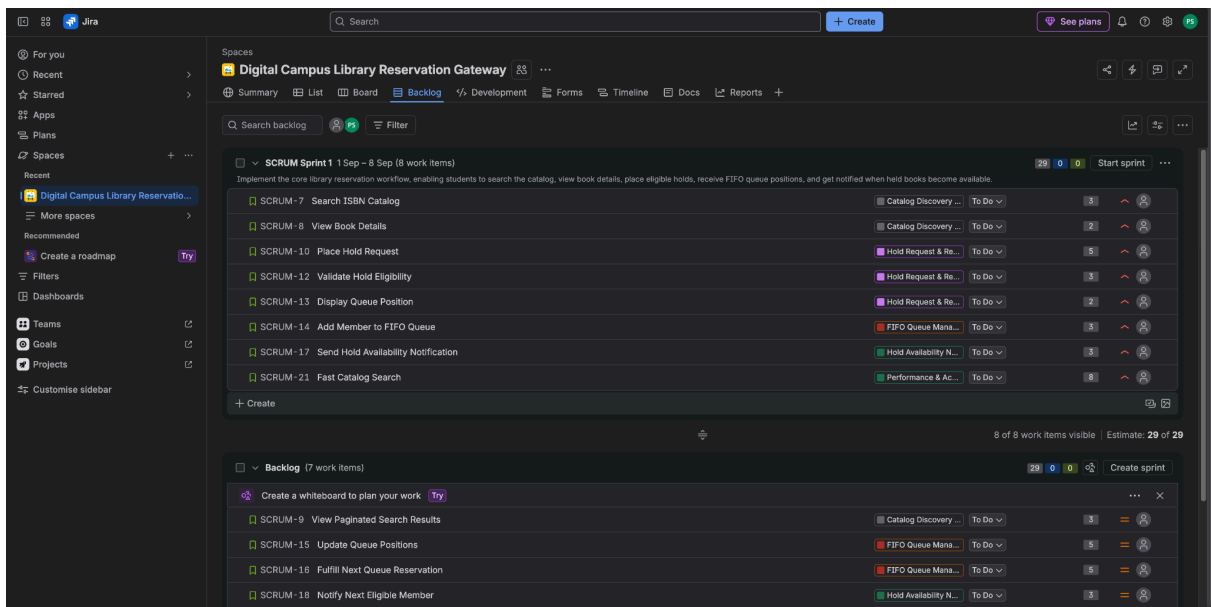
Implement the core library reservation workflow, enabling students to search the catalog, view book details, place eligible holds, receive FIFO queue positions, and get notified when held books become available.

Sprint 1 Scope

Story

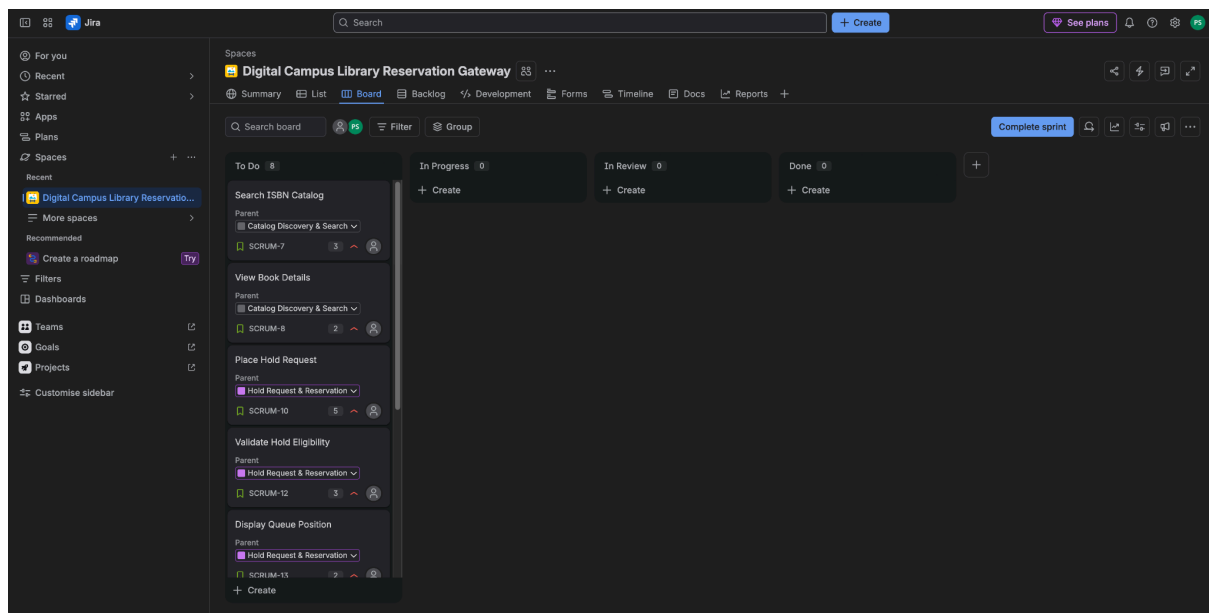
Points

Search ISBN Catalog	3
View Book Details	2
Place Hold Request	5
Validate Hold Eligibility	3
Display Queue Position	2
Add Member to FIFO Queue	3
Send Hold Availability Notification	3
Fast Catalog Search	8
Total	29



6. Sprint 1 – Execution

The Sprint 1 board was used to simulate the progress of the selected User Stories. Work items were moved through the workflow from **To Do** → **In Progress** → **In Review** → **Done**.



7. Sprint Comments

Comments were added to selected User Stories during the mock sprint to document development progress, validation, and completion.

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Attachments

Subtasks

Linked work items

Activity

In Progress

Details

Assignee

Parent

Due date

Labels

Team

Start date

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Story point estimate

Reporter

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Subtasks

Linked work items

Activity

In Progress

Details

Assignee

Parent

Due date

Labels

Team

Start date

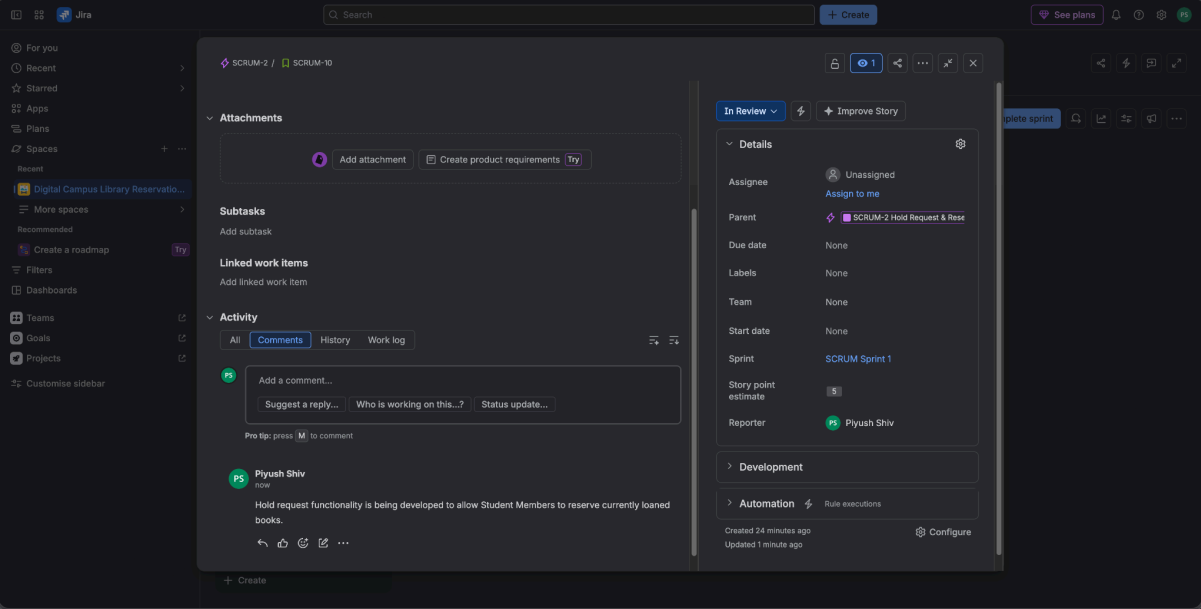
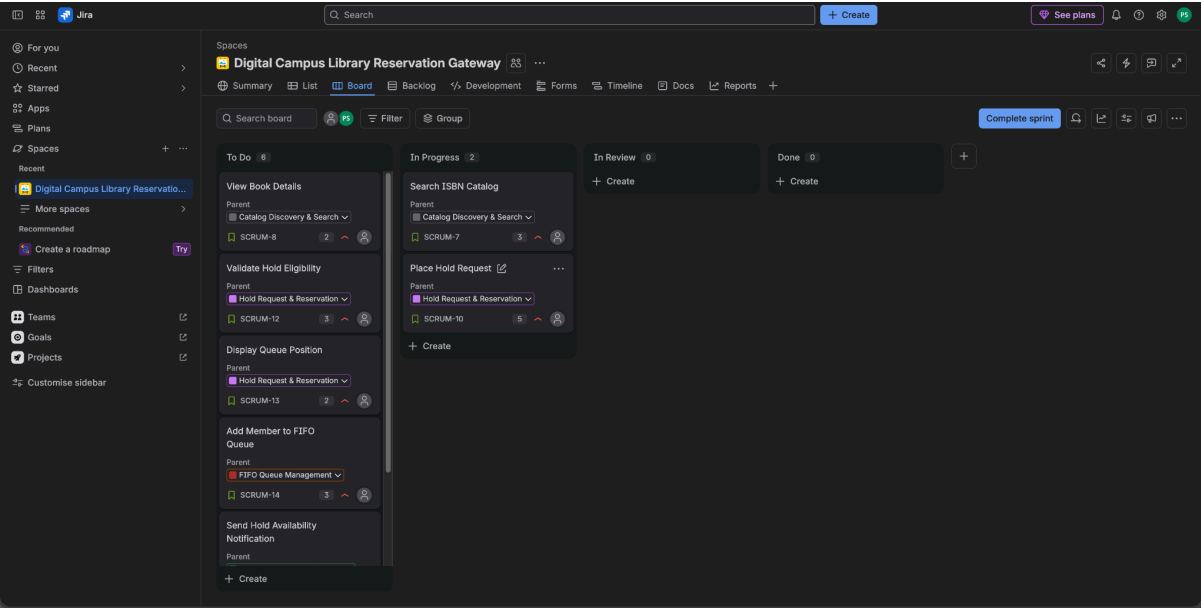
Sprint

Story point estimate

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SCRUM-1 / SCRUM-7

Add subtask

Linked work items

Add linked work item

Activity

AllCommentsHistoryWork log

Add a comment...

Suggest a reply...Who is working on this...?Status update...

Pro tip: press **M** to comment

Piyush Shiv

now

Catalog search implementation is complete. ISBN and title searches have been tested with sample library records and are ready for review.

Piyush Shiv

3 minutes ago

Development started on catalog search. ISBN and title-based search functionality is being implemented and tested against the library catalog requirements.

In Review

Improve Story

Details

Assignee

Unassigned

Assign to me

Parent

SCRUM-1 Catalog Discovery &

Due date

None

Labels

None

Team

None

Start date

None

Sprint

SCRUM Sprint 1

Story point estimate

Reporter

Piyush Shiv

Development

Automation

Rule executions

Created 30 minutes ago

Updated 2 minutes ago

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SCRUM-2 / SCRUM-10

Subtasks

Add subtask

Linked work items

Add linked work item

Activity

AllCommentsHistoryWork log

Add a comment...

Suggest a reply...Who is working on this...?Status update...

Pro tip: press **M** to comment

Piyush Shiv

now

Hold requests are now being created successfully for eligible Student Members and are ready for validation.

Piyush Shiv

1 minute ago

Hold request functionality is being developed to allow Student Members to reserve currently loaned books.

In Review

Improve Story

Details

Assignee

Unassigned

Assign to me

Parent

SCRUM-2 Hold Request & Rese

Due date

None

Labels

None

Team

None

Start date

None

Sprint

SCRUM Sprint 1

Story point estimate

Reporter

Piyush Shiv

Development

Automation

Rule executions

Created 25 minutes ago

Updated 1 minute ago

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SCRUM-1 / SCRUM-7

Done 1

Done

Improve Story

Linked work items

Add linked work item

Activity

All

Comments

History

Work log

PS

Add a comment...

Suggest a reply...

Who is working on this...?

Status update...

Pro tip: press **M** to comment

PS

Piyush Shiv

now

Catalog search has been reviewed and verified successfully. The story meets the expected search functionality.

PS

Piyush Shiv

1 minute ago

Catalog search implementation is complete. ISBN and title searches have been tested with sample library records and are ready for review.

PS

Piyush Shiv

4 minutes ago

Details

Assignee

Unassigned

Assign to me

Parent

SCRUM-1 Catalog Discovery &

Due date

None

Labels

None

Team

None

Start date

None

Sprint

SCRUM Sprint 1

Story point estimate

3

Reporter

PS Piyush Shiv

Development

Automation

Rule executions

Created 31 minutes ago

Updated 12 seconds ago

Resolved 20 seconds ago

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SCRUM-2 / SCRUM-10

Done 1

Done

Improve Story

Activity

All

Comments

History

Work log

PS

Add a comment...

Suggest a reply...

Who is working on this...?

Status update...

Pro tip: press **M** to comment

PS

Piyush Shiv

now

Hold request functionality has been tested and verified successfully.

PS

Piyush Shiv

2 minutes ago

Hold requests are now being created successfully for eligible Student Members and are ready for validation.

PS

Piyush Shiv

2 minutes ago

Hold request functionality is being developed to allow Student Members to reserve currently loaned books.

Details

Assignee

Unassigned

Assign to me

Parent

SCRUM-2 Hold Request & Rese

Due date

None

Labels

None

Team

None

Start date

None

Sprint

SCRUM Sprint 1

Story point estimate

5

Reporter

PS Piyush Shiv

Development

Automation

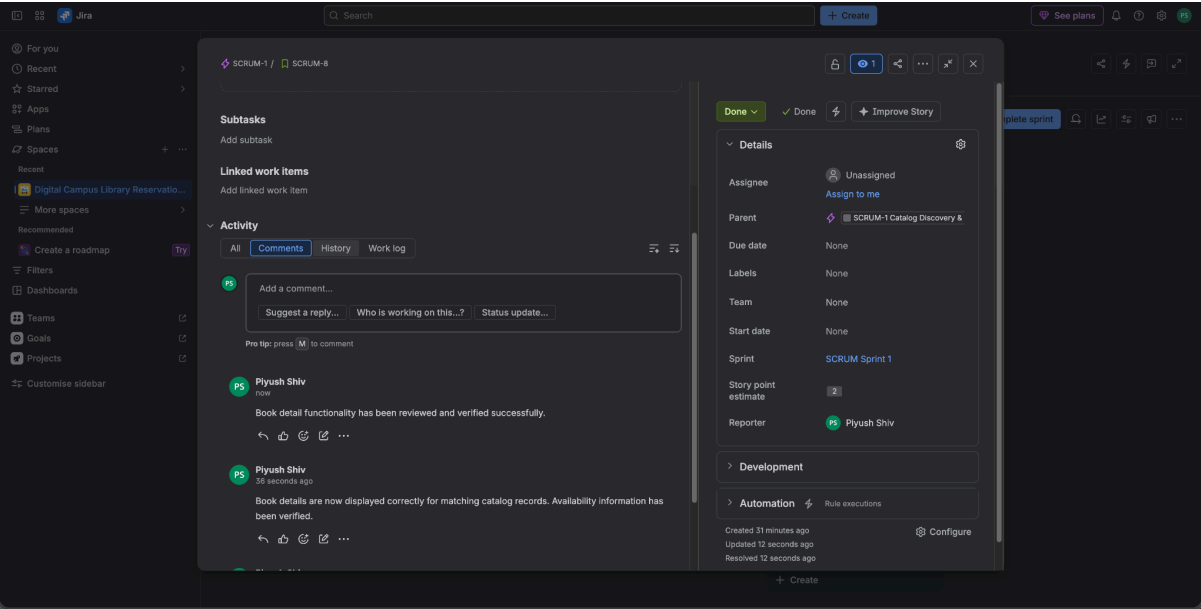
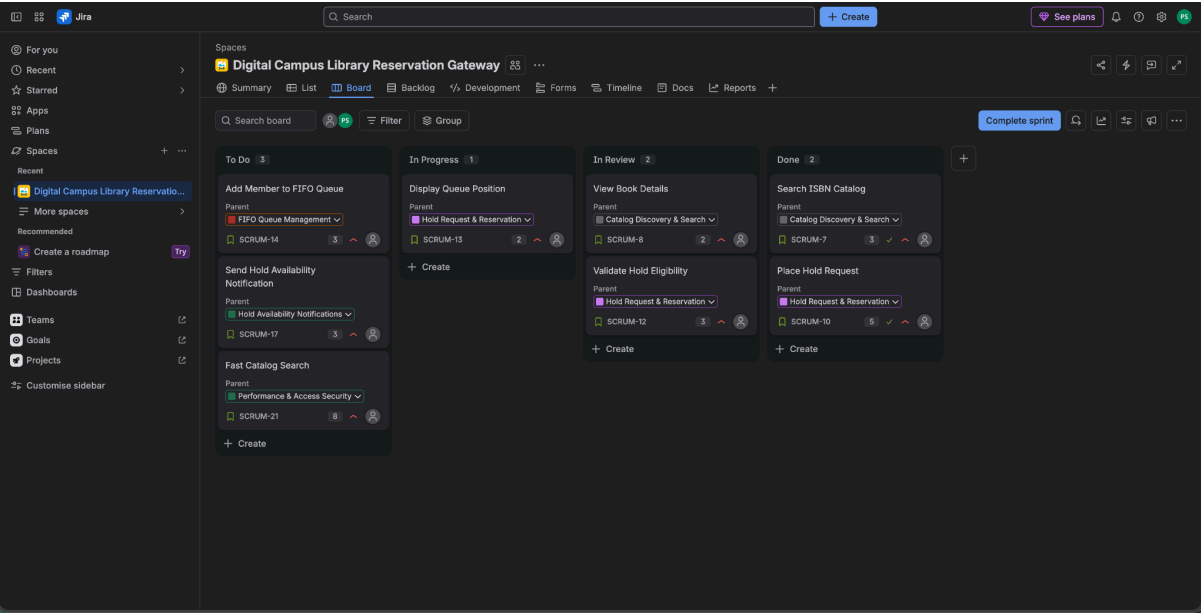
Rule executions

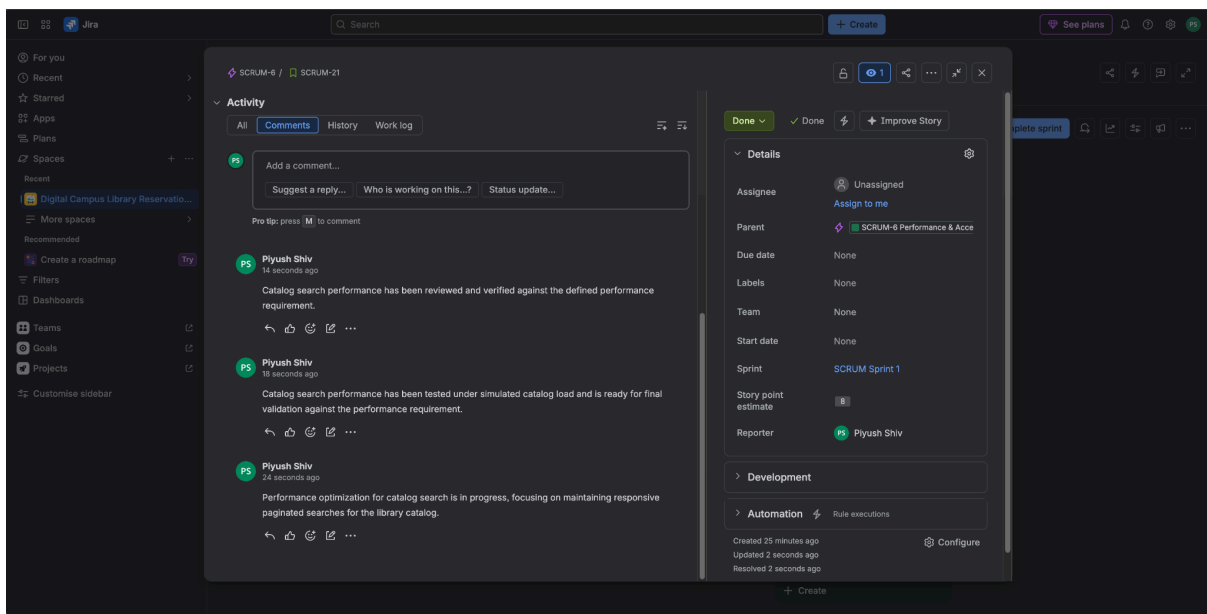
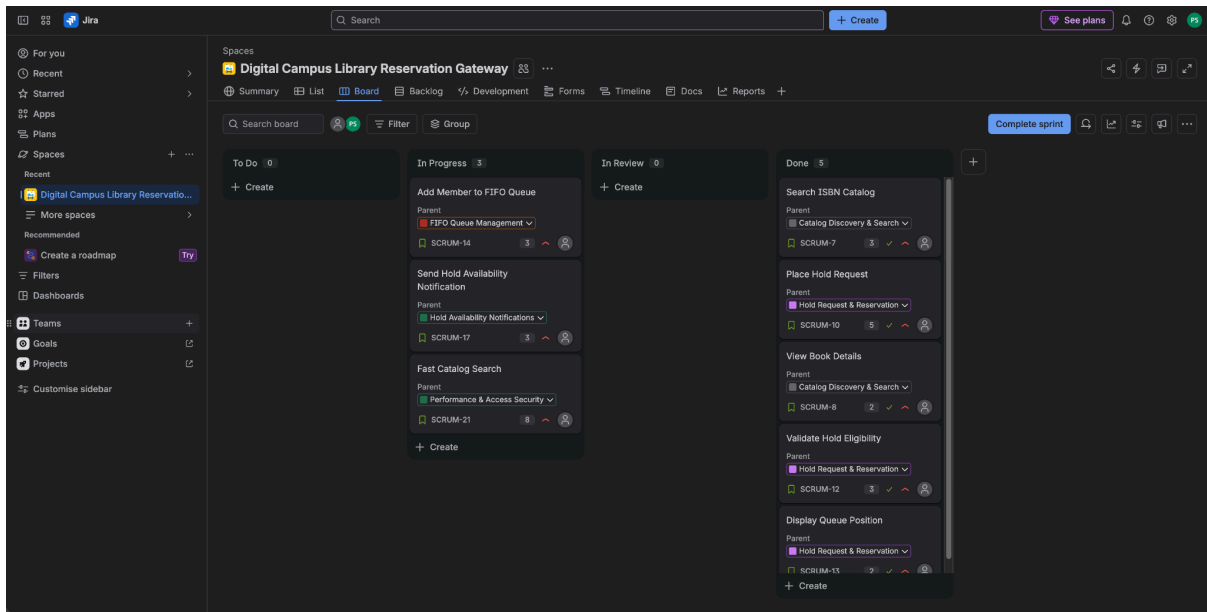
Created 27 minutes ago

Updated 2 seconds ago

Resolved 2 seconds ago

Configure





8. Sprint 1 – Burndown and Completion

The Sprint 1 progress and burndown views were used to visualize the remaining work during the sprint. The sprint was subsequently completed in Jira.

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SummaryBoardBacklogDevelopmentFormsTimelineDocsReports

Search boardFilterGroup

Complete sprint

To Do0

In Progress0

In Review0

Done8

Search ISBN Catalog

Parent

Catalog Discovery & Search

SCRUM-73

Place Hold Request

Parent

Hold Request & Reservation

SCRUM-105

View Book Details

Parent

Catalog Discovery & Search

SCRUM-82

Validate Hold Eligibility

Parent

Hold Request & Reservation

SCRUM-123

Display Queue Position

Parent

Hold Request & Reservation

SCRUM-132

▼ Sprint progress



 100% done

Done

100%

In progress

0%

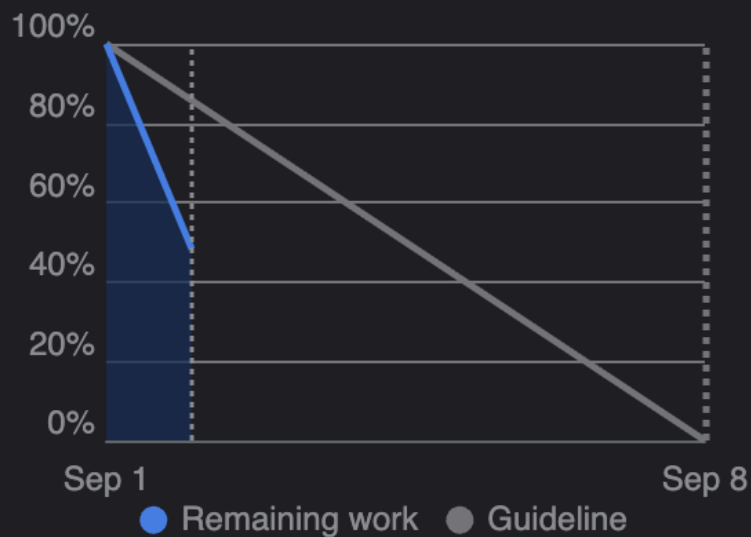
Not started

0%

▼ Sprint burndown



15 points done, 14 points to go



▼ Epic progress



This sprint is working towards **5 epics**

SCRUM-2 Hold Request & Reserva... 100% done



SCRUM-6 Performance & Access ... 100% done

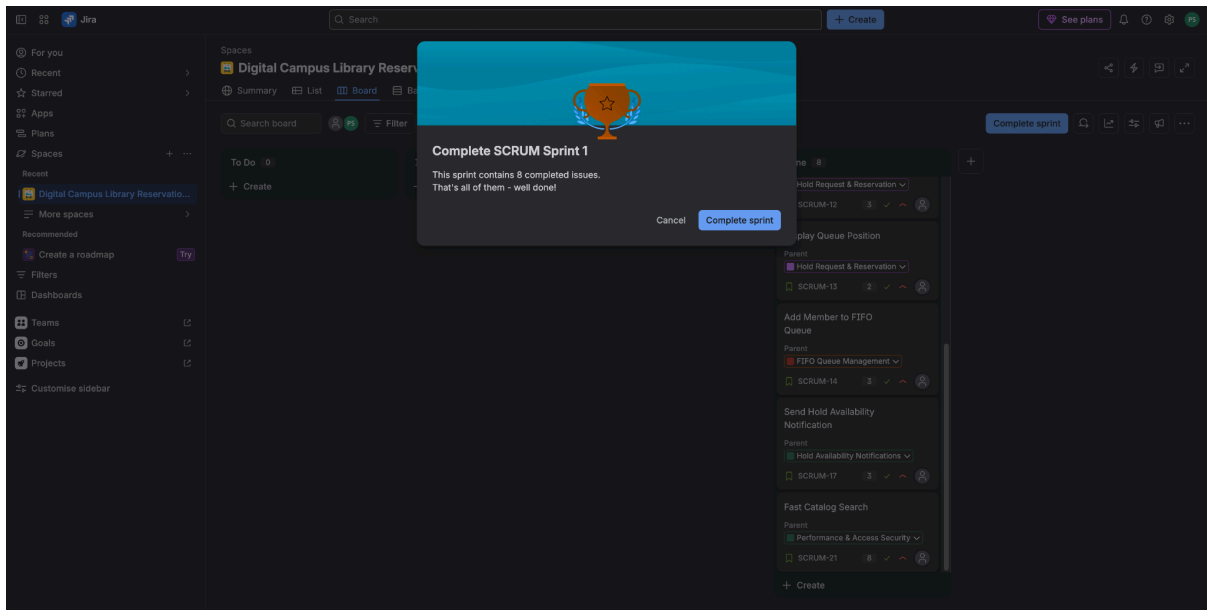


SCRUM-1 Catalog Discovery & Sea... 100% done



SCRUM-4 Hold Availability Notifica... 100% done





9. Sprint 2 – Planning

Sprint Goal

Complete the library reservation gateway by improving queue management, supporting librarian fine tracking and administration, and enforcing secure access to librarian operations.

Sprint 2 Scope

Story	Points
View Paginated Search Results	3
Update Queue Positions	5
Fulfill Next Queue Reservation	5
Notify Next Eligible Member	3
Record Overdue Fine	5
View Fine Status	3

Restrict Librarian Operations	5
Total	29

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Search backlogFilter

SCRUM Sprint 28 Sep – 15 Sep (7 work items)2900Start sprint

SCRUM-9View Paginated Search ResultsCatalog Discovery ...To Do3

SCRUM-15Update Queue PositionsFIFO Queue Mana...To Do5

SCRUM-16Fulfill Next Queue ReservationFIFO Queue Mana...To Do5

SCRUM-18Notify Next Eligible MemberHold Availability N...To Do3

SCRUM-19Record Overdue FineLibrarian Fine Man...To Do5

SCRUM-20View Fine StatusLibrarian Fine Man...To Do3

SCRUM-22Restrict Librarian OperationsPerformance & Ac...To Do5

Create

7 of 7 work items visibleEstimate: 29 of 29

Backlog(0 work items)000Create sprint

Your backlog is empty.

Create

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Search backlogFilter

SCRUM Sprint 28 Sep – 15 Sep (7 work items)2900Start sprint

Complete the library reservation gateway by improving queue management, supporting librarian fine tracking and administration, and enforcing secure access to librarian operations.

SCRUM-9View Paginated Search ResultsCatalog Discovery ...To Do3

SCRUM-15Update Queue PositionsFIFO Queue Mana...To Do5

SCRUM-16Fulfill Next Queue ReservationFIFO Queue Mana...To Do5

SCRUM-18Notify Next Eligible MemberHold Availability N...To Do3

SCRUM-19Record Overdue FineLibrarian Fine Man...To Do5

SCRUM-20View Fine StatusLibrarian Fine Man...To Do3

SCRUM-22Restrict Librarian OperationsPerformance & Ac...To Do5

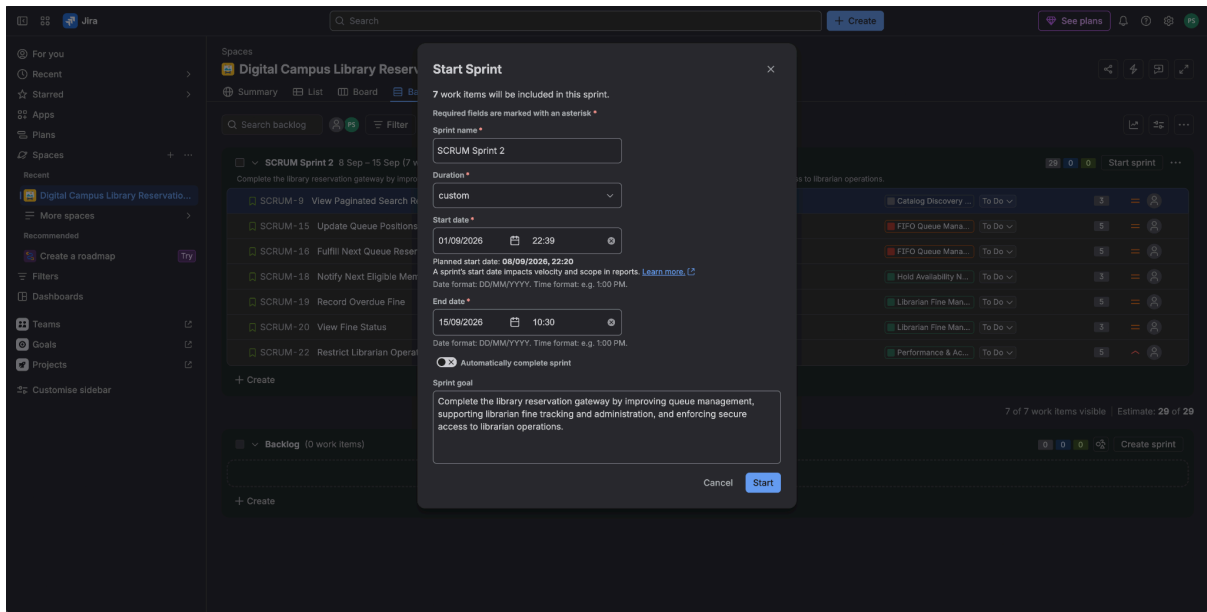
Create

7 of 7 work items visibleEstimate: 29 of 29

Backlog(0 work items)000Create sprint

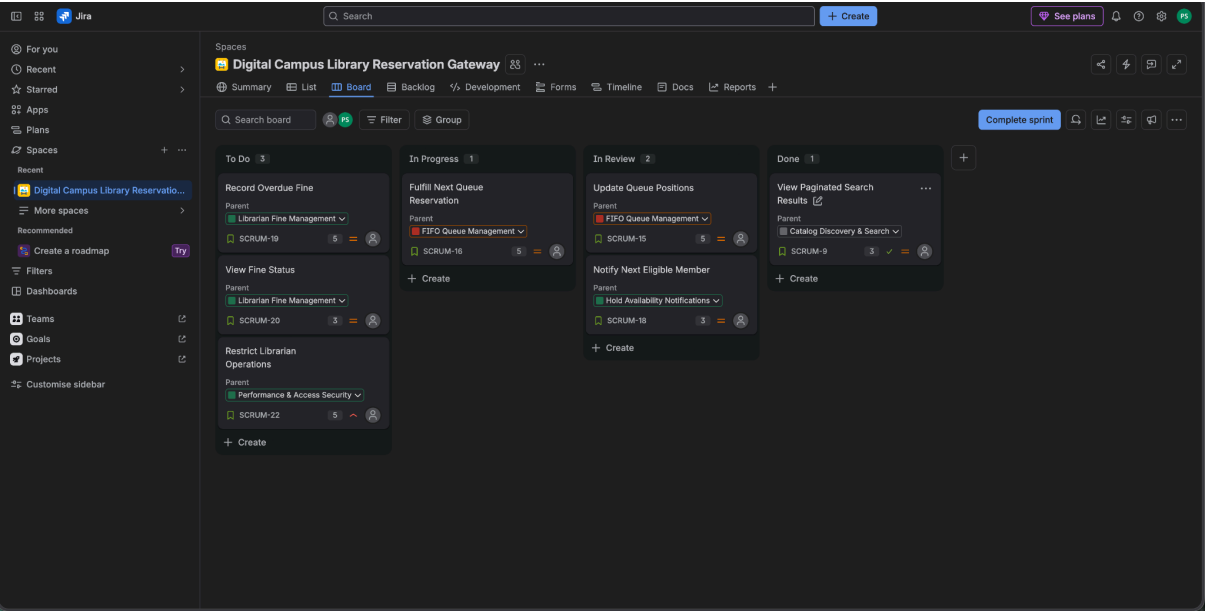
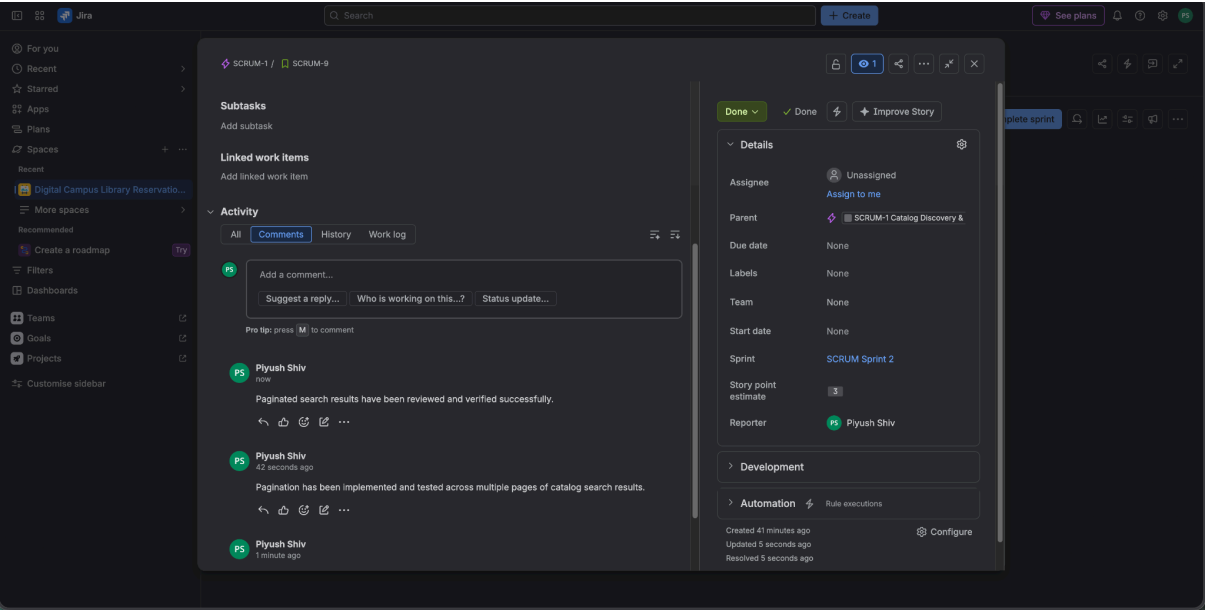
Your backlog is empty.

Create



10. Sprint 2 – Execution

The remaining User Stories were planned for Sprint 2 and progressed through the Jira workflow to simulate sprint execution.



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Search boardFilterGroup

Complete sprint

To Do1

Restrict Librarian Operations

ParentPerformance & Access Security

SCRUM-225

Create

In Progress2

Record Overdue Fine

ParentLibrarian Fine Management

SCRUM-195

View Fine Status

ParentLibrarian Fine Management

SCRUM-205

Create

In Review1

Fulfill Next Queue Reservation

ParentFIFO Queue Management

SCRUM-185

Create

Done3

View Paginated Search Results

ParentCatalog Discovery & Search

SCRUM-93

Update Queue Positions

ParentFIFO Queue Management

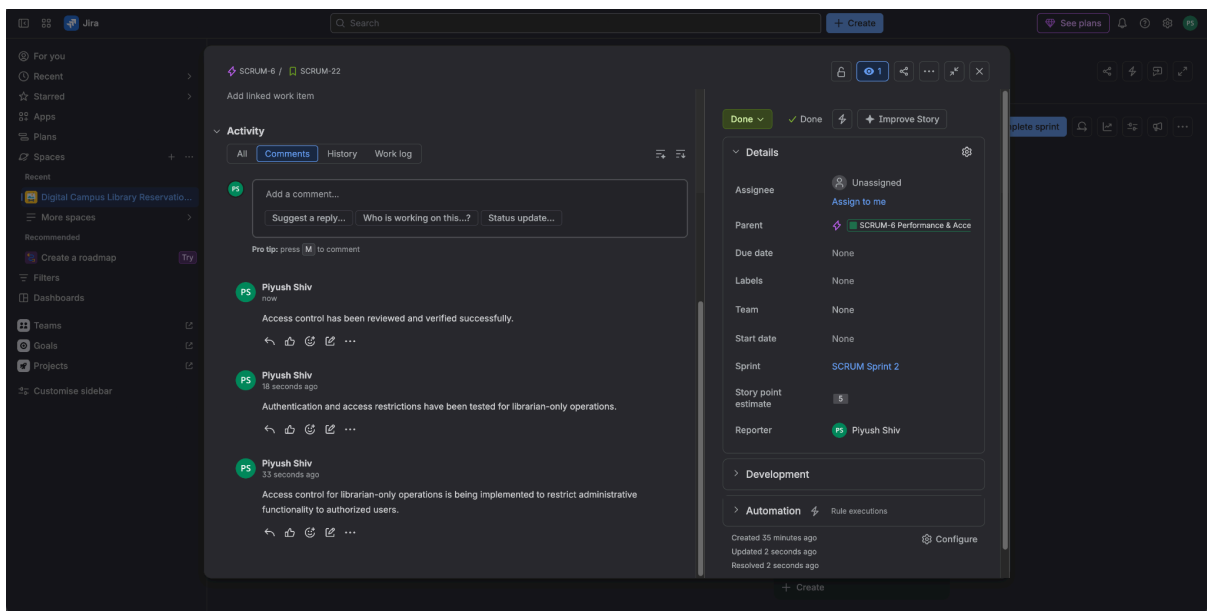
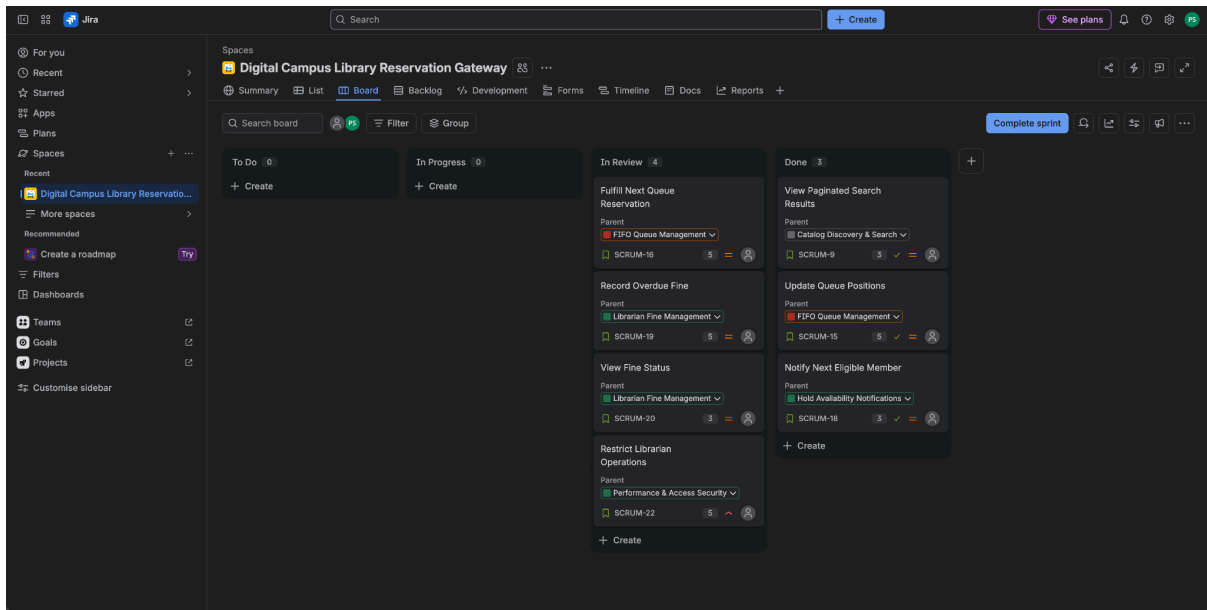
SCRUM-155

Notify Next Eligible Member

ParentHold Availability Notifications

SCRUM-183

Create



11. Sprint 2 – Burndown and Completion

The Sprint 2 progress and burndown views were used to monitor sprint work and visualize the change in remaining effort.

Jira

Q Search

+ Create

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Summary

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Backlog

Development

Forms

Timeline

Docs

Reports

Q Search board

Filter

Group

Complete sprint

To Do

+ Create

In Progress

+ Create

In Review

+ Create

Done

Hold Availability Notifications

SCRUM-18

3

Fulfill Next Queue Reservation

Parent

FIFO Queue Management

SCRUM-18

5

Record Overdue Fine

Parent

Librarian Fine Management

SCRUM-19

5

View Fine Status

Parent

Librarian Fine Management

SCRUM-20

3

Restrict Librarian Operations

Parent

Performance & Access Security

SCRUM-22

5

+ Create

https://piyushshiv26.atlassian.net/browse/SCRUM-16

▼ Sprint progress



Done	In progress	Not started
100%	0%	0%

▼ Sprint burndown



6 points done, 23 points to go



▼ Epic progress



This sprint is working towards **5 epics**

SCRUM-3 FIFO Queue Management 100% done



SCRUM-5 Librarian Fine Managem... 100% done



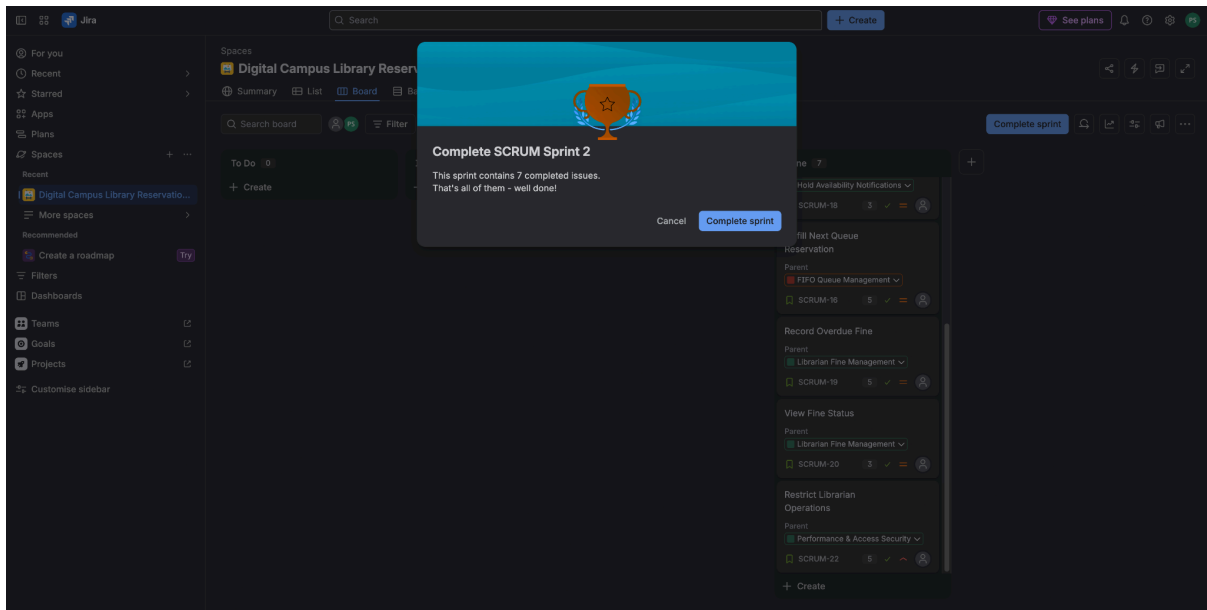
SCRUM-6 Performance & Access ... 100% done



SCRUM-4 Hold Availability Notifica... 100% done

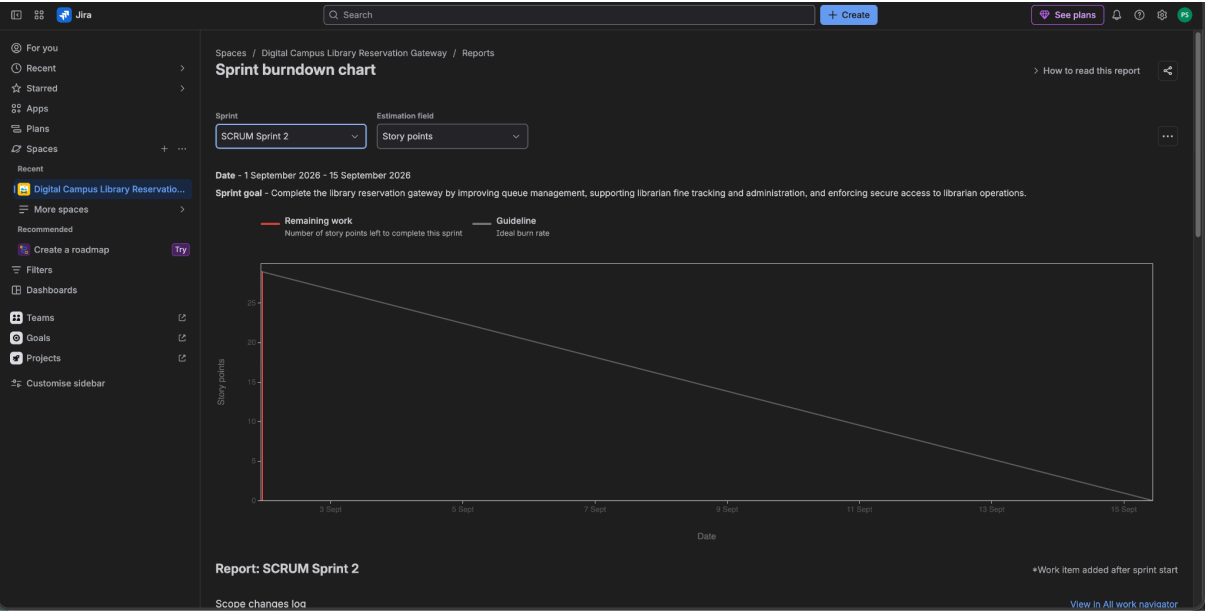
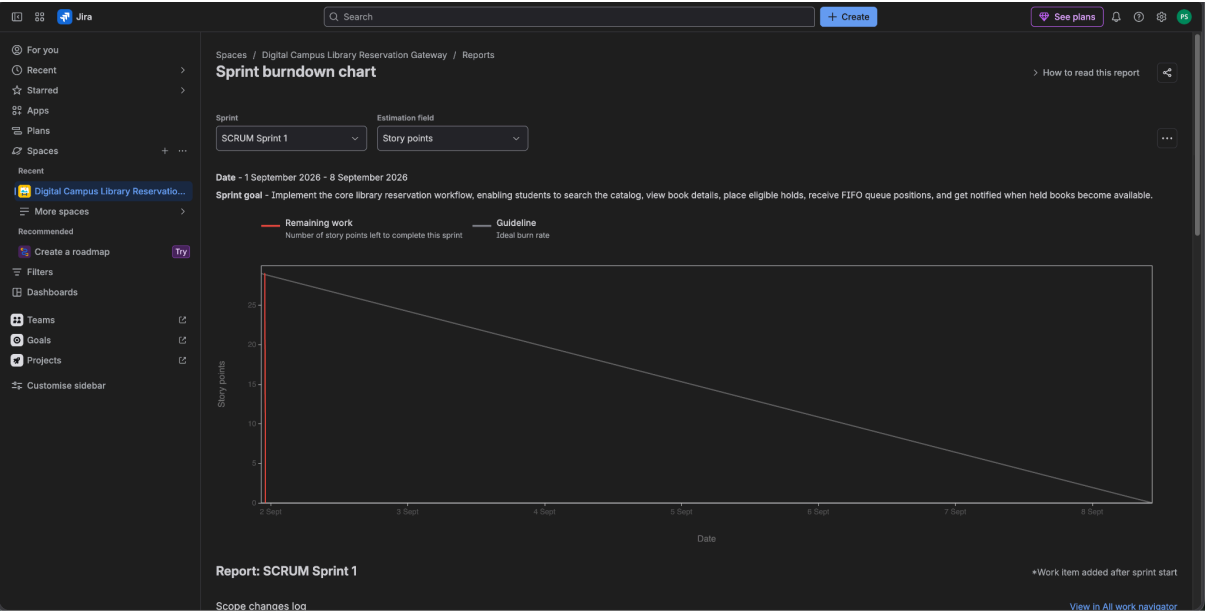


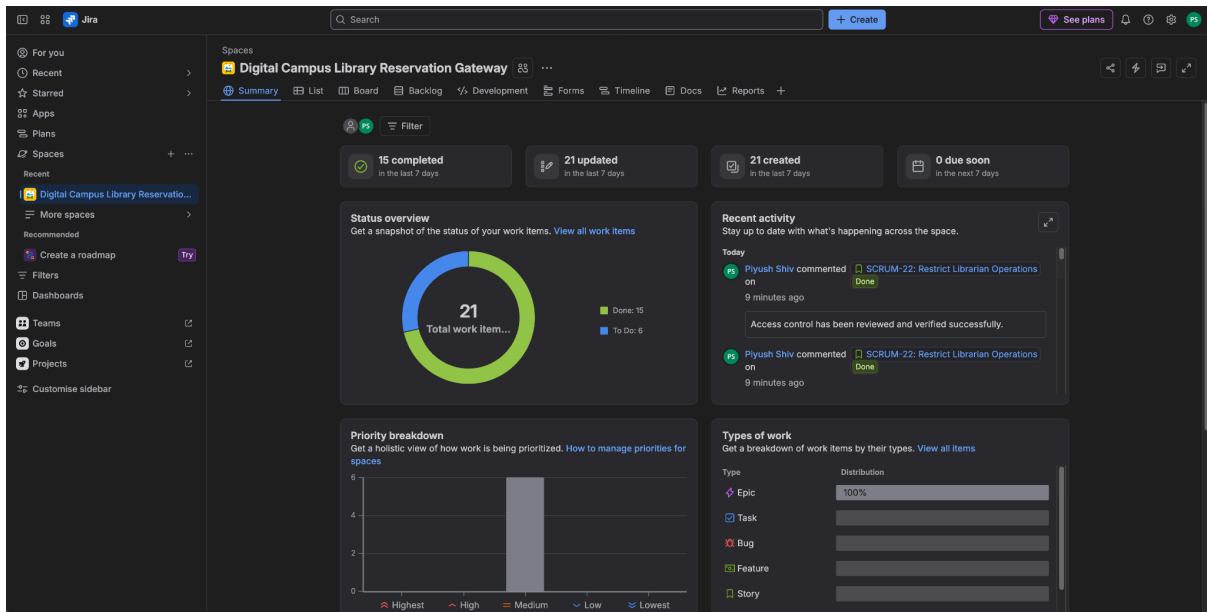
Other epics 100% done



12. Jira Reports

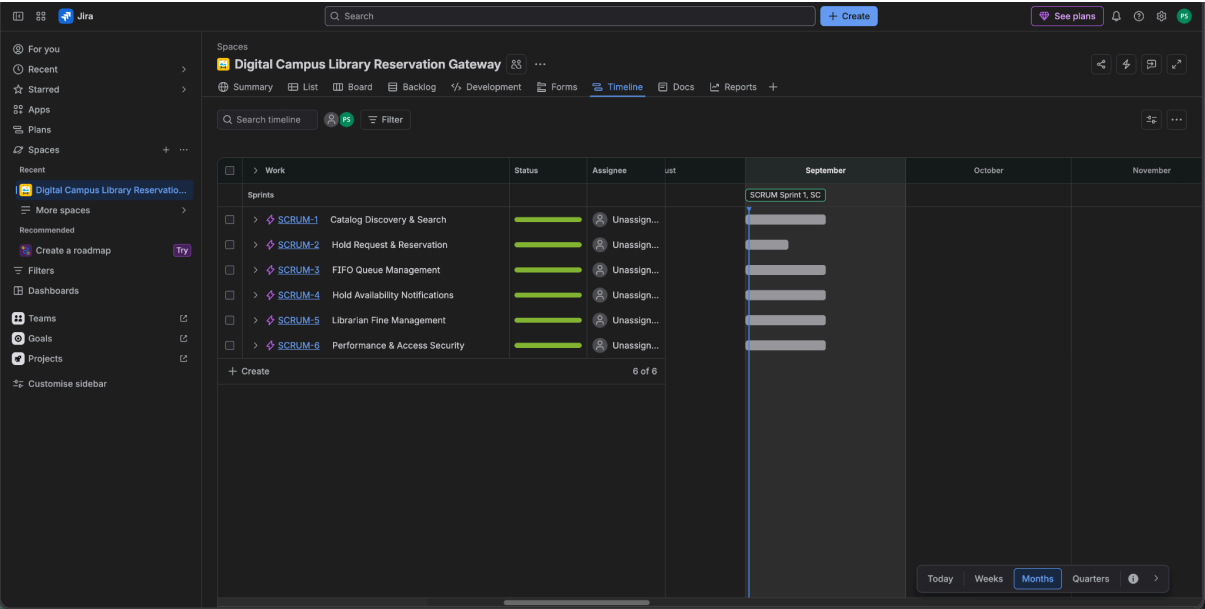
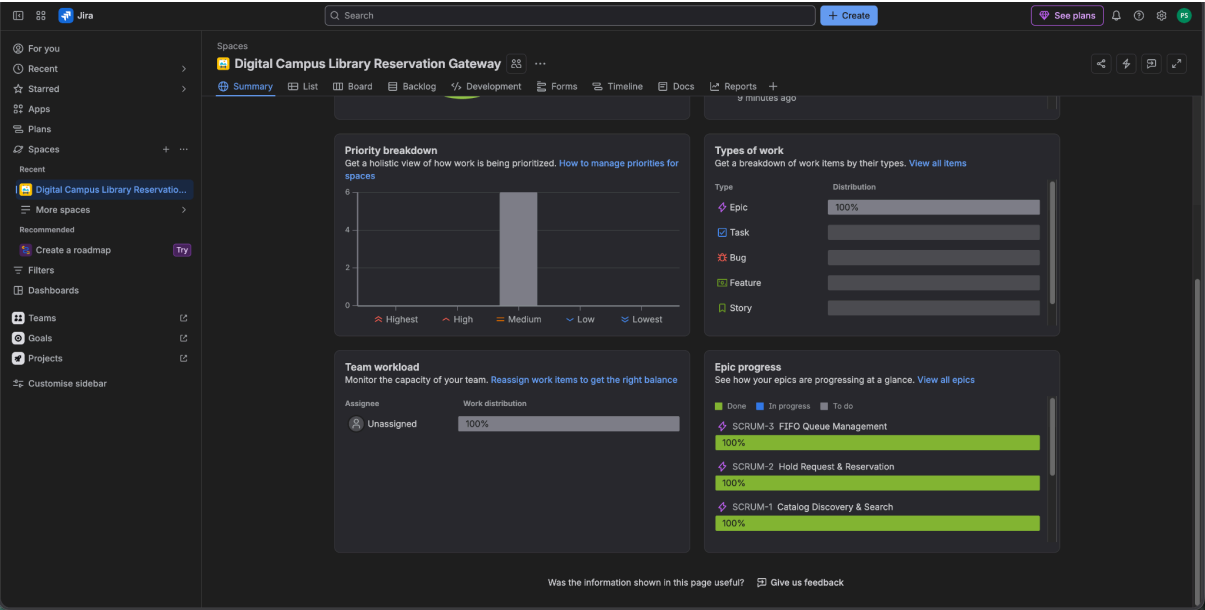
Jira reports were used to visualize sprint performance and overall project progress. The generated reports provide evidence of sprint execution, work distribution, and progress across the project.

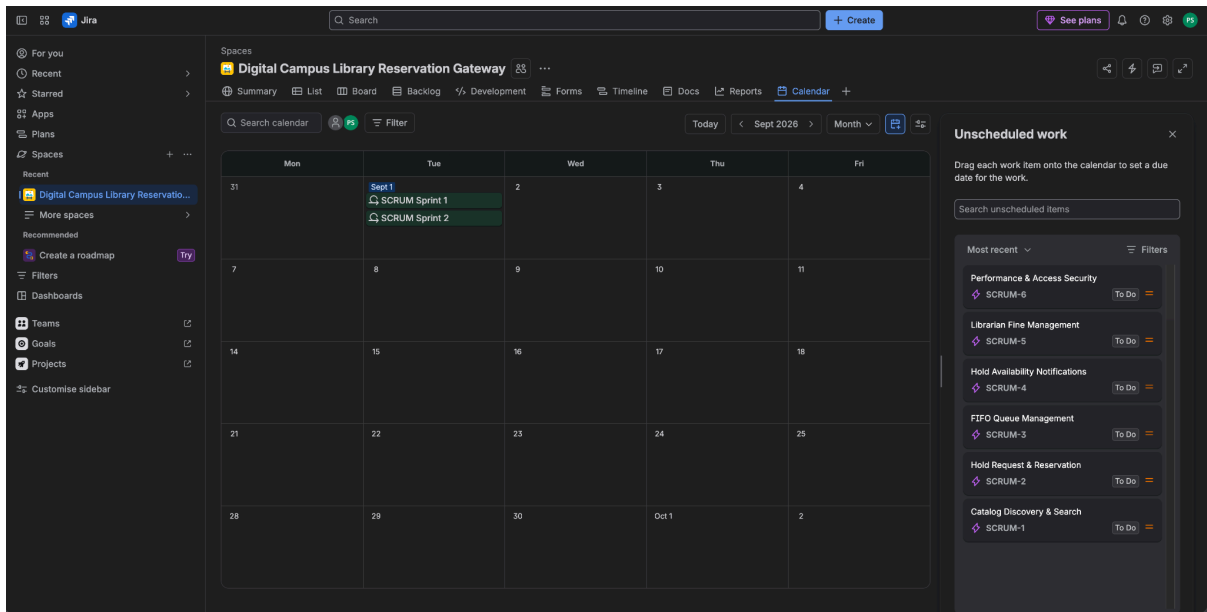




13. Project Timeline and Work Overview

The Jira Timeline and work overview were used to visualize the project's planned work items and their organization across the project.





	Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated
☐	> SCRUM-1 Catalog Discovery & Search	Unassigned	Piyush Shiv	Medium	To Do	Unresolved	01 Sept 2026, 21:49	01 Sept 2026
☐	> SCRUM-2 Hold Request & Reservation	Unassigned	Piyush Shiv	Medium	To Do	Unresolved	01 Sept 2026, 21:49	01 Sept 2026
☐	> SCRUM-3 FIFO Queue Management	Unassigned	Piyush Shiv	Medium	To Do	Unresolved	01 Sept 2026, 21:50	01 Sept 2026
☐	> SCRUM-4 Hold Availability Notifications	Unassigned	Piyush Shiv	Medium	To Do	Unresolved	01 Sept 2026, 21:50	01 Sept 2026
☐	> SCRUM-5 Librarian Fine Management	Unassigned	Piyush Shiv	Medium	To Do	Unresolved	01 Sept 2026, 21:50	01 Sept 2026
☐	> SCRUM-6 Performance & Access Security	Unassigned	Piyush Shiv	Medium	To Do	Unresolved	01 Sept 2026, 21:51	01 Sept 2026

14. Sprint Retrospective

1. What went well?

The project was successfully divided into six Epics and fifteen User Stories, making the Digital Campus Library Reservation Gateway easier to organize and manage. Priorities and Story Points helped structure the backlog, while the Scrum board provided a clear view of work progressing through To Do, In Progress, In Review, and Done.

2. What was challenging?

Estimating Story Points was challenging for features involving multiple business rules, such as FIFO queue management, hold eligibility, notification handling, and librarian access control. Some features required more effort because they involved interactions between different parts of the library reservation workflow.

3. What did the burndown chart show?

The burndown charts provided a visual representation of the remaining sprint work as User Stories progressed through the Jira workflow. They helped demonstrate how sprint progress and remaining effort could be monitored during the simulated sprints.

4. What could be improved?

Future sprint planning could use historical velocity and previous sprint performance to improve estimation accuracy. Complex features could also be broken into smaller User Stories where appropriate to reduce uncertainty and make sprint planning more predictable.

5. What was learned?

The exercise demonstrated how Agile Scrum can be applied using Jira to organize software development. It provided practical experience in creating Epics and User Stories, assigning priorities and Story Points, planning sprints, tracking work using a Scrum board, and monitoring progress through burndown charts and reports.

15. Conclusion

The Digital Campus Library Reservation Gateway was successfully organized and managed using Jira and Agile Scrum practices. The project was decomposed into six Epics and fifteen User Stories, with priorities and Story Point estimates assigned to each story. Two sprints were planned and executed using the Jira Scrum board, with User Stories progressed through To Do, In Progress, In Review, and Done states. Sprint comments, burndown charts, reports, and the project timeline were used to demonstrate Agile project tracking and monitoring. The exercise also provided practical experience in backlog management, sprint planning, estimation, and iterative progress tracking.

Tools Used

- Jira
- Agile Scrum
- Scrum Board
- Backlog
- Burndown Chart
- Jira Reports